

Research Methods — Module 06: Learning — Study Questions

1. What is computational psychiatry? How does it differ from descriptive psychiatry (DSM)?
2. How does Active Inference reframe mental disorders as computational dysfunctions?
3. What are the four main types of dysfunction (precision, structure, preferences, updating)?
4. How does the aberrant precision account explain hallucinations in schizophrenia?
5. How does overly precise sensory processing relate to delusions?
6. What is the Active Inference model of depression? How do pessimistic priors manifest?
7. How does anhedonia relate to reduced precision on positive prediction errors?
8. How does anxiety relate to elevated precision on threat prediction errors?
9. What is the Active Inference account of autism? How does altered precision hierarchy manifest?
10. Describe the 5-step translational pipeline from computational theory to treatment.
11. What is pharmacological validation? How does dopamine relate to precision?
12. How might serotonin modulate prior beliefs in Active Inference terms?
13. What is enrichment in clinical trials? How does computational phenotyping enable it?
14. What are Bayesian adaptive clinical trials? How do they use Active Inference principles?
15. What is digital phenotyping? How can smartphone tasks measure computational parameters?
16. How would you computationally distinguish two patients with "depression" who have different underlying dysfunctions?
17. What ethical issues arise from computational phenotyping in psychiatry?
18. How does this framework help explain treatment resistance?
19. How does comorbidity (multiple disorders) look through the computational lens?
20. Critically evaluate: Is computational psychiatry ready to improve patient outcomes, or is it still premature?